

CUSTOMER SHOPPING DATASET

An Interactive Data Visualization Report on Consumer Purchase Behavior

Subject: Data Visualization | SDG 8 – Decent Work & Economic Growth

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Dataset Overview

This report analyzes consumer purchase behavior using a Customer Shopping Dataset from Istanbul, Turkey. The dataset covers 99K transactions across 10 shopping malls (2021–2023), with data on customer demographics, product categories, payment methods, and sales performance.

Key Performance Indicators

69M	689	299K	99K
Total Sales	Avg Sales	Total Quantity	Total Transactions

Customer & Product Analysis

1. Customer Analysis

This section examines how different customer segments — by age group and gender — contribute to overall sales performance.

Sales by Age Group

Age Group	Total Sales
Senior	33M
Adult	27M
Young	9M

Senior customers generate the highest revenue (33M), followed by Adults (27M) and Young shoppers (9M).

Gender Distribution & Sales

Gender	Share of Sales
Female	59.71% (41M)
Male	40.29% (28M)

Female customers contribute a significantly higher share of total sales (59.71%) compared to male customers (40.29%).

Key Insight: Sales from adult customers remain stable across months, while younger segments show lower and fluctuating trends. Purchasing frequency is highest in the Senior group (143K units), followed by Adults (116K) and Young shoppers (41K).

2. Product Analysis

This section investigates which product categories drive revenue, demand, and distribution across shopping malls.

Insight	Finding
Top Revenue Category	Clothing (31M)
Second Highest	Technology (18M)
Highest Demand (Quantity)	Clothing – dominates year-round
Seasonal Peak	Books & Clothing spike in Jan–Feb
Stable Categories	Cosmetics, Food & Beverage
Lowest Revenue	Toys & Souvenir

Payment, Location & Time Analysis

3. Payment Method Analysis

Examines how payment method contributions vary over time and their impact on overall sales and profit.

Payment Method	Total Sales (₹)	Total Profit (₹)
Cash	30,705,031	2,25,66,448.60
Credit Card	24,051,477	1,76,15,424.75
Debit Card	13,794,858	1,01,19,285.49
Total	68,551,366	5,03,01,158.85

Digital payment methods (Cash + Credit Card) consistently contribute the largest share of monthly sales. Total transaction count stands at 99.46K, with cash being the most preferred payment method, followed by credit and debit cards.

4. Location Analysis

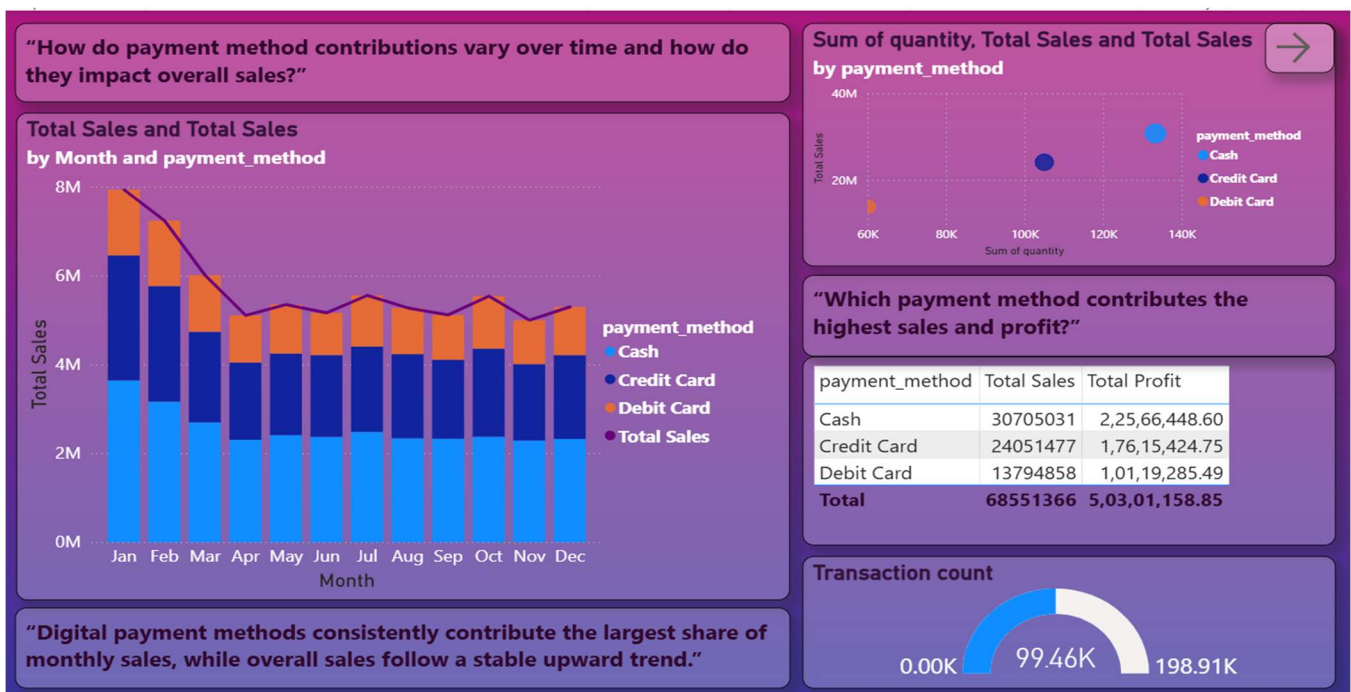
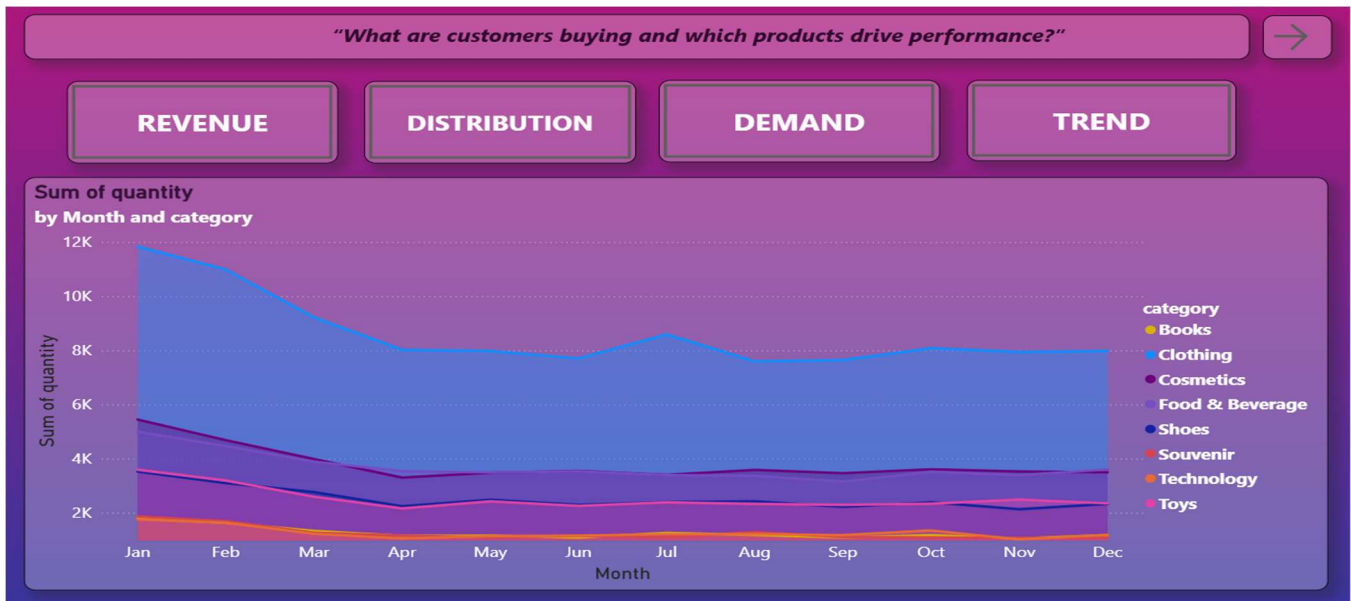
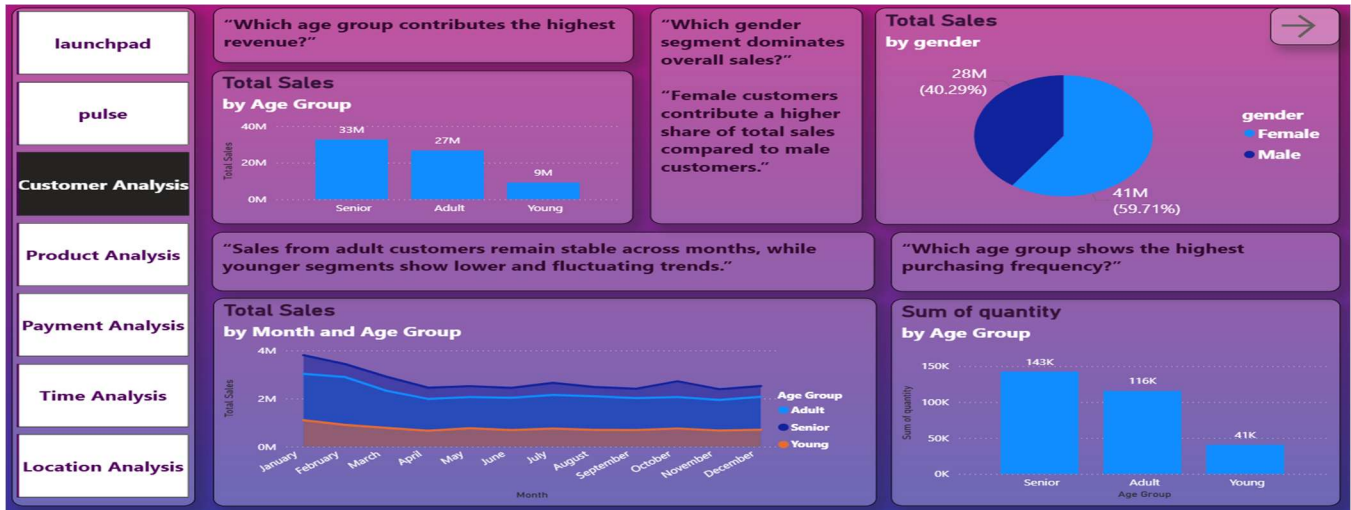
Explores how shopping mall location influences sales performance and category distribution across Istanbul.

Shopping Mall	Total Sales (₹)
Mall of Istanbul	13,851,738
Kanyon	13,710,755
Istinye Park	6,717,078
Metrocity	~13.9M
Cevahir AVM	3,433,672
Emaar Square Mall	3,390,408
Forum Istanbul	3,336,074

Business performance is strongly influenced by shopping mall location. Mall of Istanbul and Kanyon are the top-performing malls (each exceeding 13.7M in sales). Metrocity also achieves approximately 13.9M in sales. Cevahir AVM, Emaar Square Mall, and Forum Istanbul are lower performers, suggesting opportunities for targeted marketing campaigns.

Screenshot of Dashboard

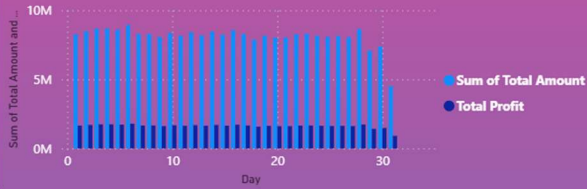




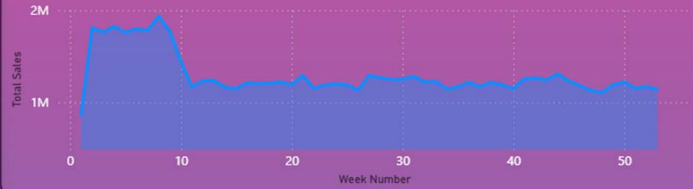
"How does business performance change over time and what growth patterns can be observed?"



Sum of Total Amount and Total Profit
by Day



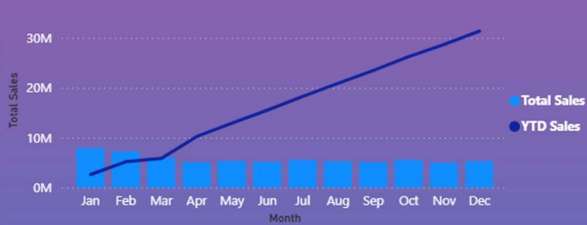
Total Sales
by Week Number



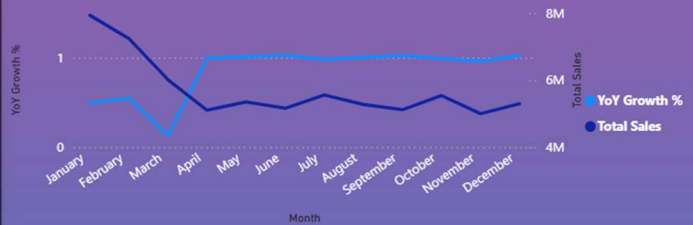
"Consistent monthly performance drives strong cumulative yearly growth."

Growth is inconsistent despite stable sales, highlighting variation in year-over-year performance."

Total Sales and YTD Sales
by Month



YoY Growth % and Total Sales
by Month



"Where does the business perform best and how does location influence sales?"

shopping_...

- ☐ Cevahir AVM
- ☐ Emaar Squ...
- ☐ Forum Ista...
- ☐ Istinye Park
- ☐ Kanyon
- ☐ Mall of Ista..
- ☐ Metrocity
- ☐ Metropol ...
- ☐ Viaport Ou...

Total Sales
by shopping_mall



Total Sales
by shopping_mall



"This analysis highlights that business performance is strongly influenced by shopping mall locations, with certain malls contributing significantly more revenue."

Count of category
by shopping_mall



Sales per Customer and Total Sales
by shopping_mall

